

## **1.Problem Statement and Motivation**

- Business problem: IT help desk. As technology is a staple in almost every business, naturally this comes with the unwanted side effects and issues associated with that technology, whether it be technological or human error.
- Target users: anyone within the business faced with technological issues, so essentially everyone in the business at some point.
- Why Agentic AI: An agentic AI solution is necessary for this issue, because each user may be experiencing different and nuanced problems that need tailored solutions. A generative AI solution may not be able to plan appropriately for user issues, especially if their problems require a follow up later or human escalation.

## **2.Use Cases and Scenarios**

- Scenario: A user wants to fix the issue they are having with their work laptop. They are having trouble connecting to the work home portal.
- Steps: The agent will take this information, ask questions that will give more context and potentially give the user a solution. For example, “Is your work issued VPN active?”. In this scenario the user would check their VPN, see it is inactive and activate it, solving their issue easily and quickly. Then, the agent will ask if any of these items have solved the issue, and if not, the agent will use the newfound information to recommend more solutions.
- Scenario: A user wants to fix the issue they are having with a corrupted word document that contains a deliverable for their boss.

- Steps: The agent will take this information and ask questions such as, “did you upload the document to the cloud?”. This will push the user in the right direction to redownload the file from the cloud to replace the corrupted version on their local storage. If there is no file in cloud storage, the agent will recommend attempting to use a file restoration service for the specific file type, and check if this works. If not, the agent will escalate to a human, and it will also advise the user to upload all documents to the cloud next time, and ensure they are frequently updated.
- Scenario: A new hire wants to gain access to company issued software required to perform their job duties.
- Steps: The agent will ask how long it’s been since they were added to the system. If it has been abnormally long, the agent will give the user a list of tasks to attempt to receive the software. If this still does not work, the agent will automatically put in a ticket to the IT and HR team, as the issue could be on either end. The agent will also send a copy of this ticket to the user’s boss.

### **3.Initial System Design**

List of agents:

- Intake Agent – collect issues and details, sort through relevant information
- Classification Agent – determine the type of issue being faced and diagnostic type
- Diagnostic Agent – propose fixes, ask any follow-up questions based on user issues.  
Escalate to a human if it cannot form any more fixes or if they become too complex
- Tool Agent – for API calls if password resets, verification checks, etc. are needed

- Critic Agent – validates solutions and ensures they are safe and clear

Expected tools and Data Sources:

- OpenAI API, Password reset API, Ticket creation and status check API
- Internal knowledge base – list of troubleshooting guides and common issues put together by the company IT support team, as well as previous tickets. The agents will use this to check if the user issue has been dealt with before.
- SQLite database to log resolved issues

#### **4. Feasibility and Risks**

Potential challenges

- Hallucination which is to be mitigated by the Intake Agent. If user input is too ambiguous do not attempt to force a solution
- Unsafe solution generation which is to be mitigated by the Critic Agent

Constraints

- Model context restraints if the internal knowledge base it's pulling from is too large
- Sensitive user data exposure, so process all data locally if possible
- High resource consumption, so use sequential execution as opposed to parallel

Timeline by week

- Problem definition (1), Front-end development (2), Single-agent prototype (3), Multi-agent workflow (4-5), Tool integration and base knowledge setup (6), Testing (7), Final report and demo (8+)